

Stephen A. Rice

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PROFESSIONAL SUMMARY

Backend and systems engineer with several years of experience building reliable services in Rust, Go, and C#. Experienced designing event-driven systems, real-time data pipelines, and APIs that integrate with hardware-backed systems. Strong focus on reliability, observability, and concurrency safety, with experience deploying containerized services and debugging complex production systems.

WORK EXPERIENCE

Software Engineer - Promess Incorporated

Brighton, MI | *February 2021 - May 2025*

- Designed and implemented production backend services and **gRPC APIs in Rust and C#** to integrate industrial press controllers across customer deployments, taking ownership from design through release.
- Built **event-driven, low-latency** systems using pub/sub architectures (**MQTT, gRPC**) supporting real-time data flows (~1k updates/sec).
- Contributed to a platform redesign using **Rust** services, **GraphQL** APIs, and **WebAssembly** modules to improve real-time data delivery and system maintainability.
- Developed automated **behavior-monitoring and end-to-end testing tools in Rust** using BDD/TDD practices, surfacing defects earlier and reducing release-time issues.
- Built a backend microservice in **Go** for barcode-driven workflows, integrating with existing production systems via MQTT.
- Optimized SQL data access paths, reducing load times from several minutes to under one second for technical operators.
- Deployed and maintained backend services in **containerized Linux environments** using **Docker and Kubernetes**.
- Maintained and extended production desktop and embedded software (**C#/.NET, C++, XML**) for the company's flagship industrial control product.
- Collaborated using modern SDLC practices including **Git, CI/CD pipelines, Azure DevOps, Jira, and Confluence**.

Graduate Assistant - Eastern Michigan University

Ypsilanti, MI | *January 2019 - December 2020*

- Developed **multi-threaded web scraping and static analysis tools** in Python to mine large GitHub codebases for NLP and machine learning research.
- Trained and evaluated students as a tutor in **Java and Python**.
- Aided as a teaching assistant for **applied cryptography** coursework.

EDUCATION

Master of Science in Computer Science

Eastern Michigan University

- Focus areas: systems, machine learning, and software engineering
- Thesis: trained a large language model for punctuation prediction using **PyTorch** on **Google Cloud Platform**

Bachelor of Science in Nuclear Engineering & Radiological Sciences

University of Michigan

- Focus areas: energy systems, risk analysis, and computational modeling
- Programming experience in **C++, Python, and Matlab**

PROJECTS

Eventful - Async Event Processing Service (Rust, Tokio, Axum)

github.com/grandfoosier/eventful

- Designed an event processing service demonstrating production-style service design.
- Implemented idempotent event ingestion and retry state machines.
- Designed bounded concurrency using Tokio semaphores and worker pools.
- Added Prometheus metrics and structured tracing for production-style observability.
- Implemented backpressure handling, retry scheduling, and graceful shutdown semantics.

SKILLS

Programming Languages

Rust, Go, C#, Python, C++, SQL

Systems & APIs

REST, gRPC, MQTT, protobuf, event-driven architectures

Infrastructure

Linux, Docker, Kubernetes, CI/CD

Practices

Observability, async systems, distributed messaging, system design